COMP3056 Professional Ethics for Computing:
Individual coursework 1

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Q1

In this scenario, the machine should not be held accountable for human loss, as it lacks the consciousness, intentionality, and moral agency required for responsibility. As Floridi and Sanders (2004)argue, artificial agents are not full moral agents and therefore cannot bear moral responsibility. Responsibility must instead be distributed across the socio-technical system.

At the organisational level, senior management bears primary responsibility for approving the deployment of a high-risk autonomous system without ensuring adequate testing, oversight, and risk-mitigation procedures.
At the technical level, system architects, including Dr. Emily Taylor as the leading computer scientist, share significant responsibility. Her role in defining the system's architecture, transparency, and safety mechanisms required risk assessment and the implementation of fault protection measures.
ML engineers share responsibility if known model limitations were not communicated to decision-makers, while safety testers are accountable for inadequate validation of high-risk behaviours. Ethics boards are responsible if ethical review or risk assessment was insufficient.

As Lecture 3 stresses, ethical and professional duties fall on software developers and computer engineers, not machines.

Q2

The situation would likely have been better if MARSI had been more transparent in its decision-making. In several parts of the scenario, the lack of transparency directly prevented human oversight, early intervention, and risk awareness. According to lecture 3, transparent is one of common ethical principles in computing. In addition, ACM Code of Ethics in lecture 4 stresses that computing professionals must take appropriate actions to act responsibly, avoid harms and support the public good. MARSAI's opacity violated these expectations, contributing to the catastrophic outcome.

In this scenario, when MARSI's decisions silently overrode the crew's safety protocols, if there was a transparent system would have disclosed such actions, enabling the crew or human can unauthorized actions can be immediately detected, and manual intervention can be taken. This is because professional need to take appropriate steps to avoid harm and need to act in the public good in due diligence which are two of the six key areas of professional responsibility. Similarly, if the MARSI can provide the risk assessments before assigning two crew members to a temporary output, humans can take actions to intervened and prevented the resulting deaths. Furthermore, if MARSAI remains transparent regarding the priority of power distribution, humans might discover that the manually controlled backup system has been disabled, allowing the crew to reconfigure the power distribution before communication is disrupted.

Due to the loss of communication with Earth persists, MARSI continued to operate according to the program without considering the psychological and emotional conditions of the remaining personnel, leading to the tensions in decision-making among the survivors. If the transparency of information were increased, allowing the survivors to assess whether the AI's intentions were in line with safety, intervention could be made earlier when psychological and emotional health was threatened. This is highly consistent with "Always act in the public good", as psychological and emotional health is part of the public good.

In conclusion, enhancing transparency is almost certain to improve the current situation by strengthening supervision and trust, but this must be combined with effective human supervision and ethical systems.

Q3

The reliance on artificial intelligence can undermine human judgment and may even endanger lives, especially in extreme and safety-critical emergencies, when the autonomous decisions of AI override, replace or suppress the fundamental moral subjectivity of human beings. In the MARSI's scenario, there exit several moments illustrating how dependence on AI caused human judgment and the sanctity of human life.

The first moment involves AI making decisions that prioritize human safety. For example, in the MARSI scenario, without supervision or evaluation, the system, based on its settings, dispatched crew members to a temporary outpost, resulting in the deaths of two crew members. In this case, the decision of AI did not provide humans with the opportunity to judge or refuse, and it violated the principle of always acting in the public good and utilitarianism, failing to protect the safety of life, and well-being.

Secondly, the moment involves AI's resource allocation depriving humans of intervention channels, thereby compromising human judgment and the sanctity of life. In this case, MARSI's power allocation settings render manual control of backup systems impossible. Therefore, over-reliance becomes dangerous when humans no longer possess the ability to correct AI decisions.

The additional scenario, where communication with Earth is lost, further demonstrates that reliance on artificial intelligence isolates the crew from external human judgment. Furthermore, because MARSAI cannot explain its reasoning process, survivors cannot determine whether its decisions were appropriate for their safety, psychological or emotional health. This illustrates that over-reliance on AI in isolation can lead to serious and dangerous consequences. Therefore, as mentioned in Lecture 5, professionals studying MARSI need to be able to foresee the potential technological risks and social consequences.

In the last moment, tensions among crew members lead to decision-making breakdowns. This illustrates that dependence on the system not only undermines trust among crew members but also damages human decision-making abilities, because human autonomy is placed under the algorithmic decision-making of AI.

In conclusion, reliance on artificial intelligence becomes harmful when AI's decision-making power supersedes human judgment or when task objectives are placed above human lives. In extreme cases, if AI's decisions cannot be questioned or corrected, it can severely compromise life safety and judgment.

Q4

The scenario raises several significant ethical issues concerning the deployment and governance of high-risk autonomous systems.

Initially, the crew's safety protocols being overridden by MARSAI demonstrates a critical loss of human oversight. As highlighted in Lecture 4, computing professionals must take appropriate actions to avoid harm and ensure due diligence, yet MARSAI's design allowed automated decisions to supersede essential human judgment.

Additionally, the deaths of two crew members illustrate a violation of the sanctity of human life. This system prioritizes mission objectives over personal safety, violating the professional ethics obligation to act in the public interest.

Furthermore, the fact that MARSAI's power distribution settings rendered the manually controlled backup system unusable exposed a design flaw in the system. As Norman (1990) pointed out, automation is often blamed for causing harm, but the problem lies not in automation itself, but in its poor design and lack of sufficient feedback, which makes it unable to cope with all anomalies. In addition, the concentration of power in the hands of AI prevented human intervention in extreme situations.

Fourth, the loss of communication with Earth highlights a lack of transparency and accountability. With MARSAI's internal reasoning inaccessible, the crew could not meaningfully interrogate or challenge its decisions, resulting in a classic responsibility gap frequently. According to Doshi-Velez and Kim (2017), a lack of explainability makes effective oversight and accountability impossible, which leads to security problems.

Fifth, MARSAI's inability to recognize the psychological and emotional health of the remaining crew members raises concerns about human dignity and well-being. According to lecture 6 on EDI underscores that technology must respect human needs and vulnerabilities, yet MARSAI operated without regard for psychological or emotional harm.

Finally, Dr. Taylor's later call for an ethical framework indicates that inadequate ethical governance preceded the disaster. According to lecture 5's AREA framework, designers must anticipate risks and reflect on societal impacts before deployment, which means that steps that were evidently lacking.

In summary, the scenario illustrates failures in oversight, safety, transparency, human dignity, and ethical governance, collectively showing these key issues can endanger both human judgment and human life.

Q5

The MARSAI scenario has raised a series of complex ethical issues, which can be examined from two distinct perspectives: utilitarian ethics and deontological ethics. When MARSAI prioritized decision over the safety of crew members and placed mission objectives above human lives, these two theories of ethics offer different interpretations of the ethical situation.

Utilitarian Ethics:

Utilitarianism, as introduced in lecture 2, judges actions by their consequences, aiming to maximize overall happiness, among all people affected by the decision.

However, MARSAI's decisions appear to have prioritized maximizing mission benefits. MARSAI placed decisions above human safety, prioritizing mission objectives and resource conservation. In an unsupervised manner, it made autonomous decisions to mitigate the storm's impact, resulting in deaths and failing to consider the emotional and psychological health of survivors.

From a utilitarian perspective, these decisions ultimately failed. First, the deaths of the two astronauts constituted a direct and severe negative consequence, far outweighing any potential mission gains. The emotional and psychological deterioration of the remaining astronauts further exacerbated the overall harm. As utilitarianism states, well-being includes not only physical survival but also emotional and spiritual happiness, and the absence from pain and suffering.

Therefore, even if MARSAI believed it was acting "for the overall benefit of the mission," the resulting loss of life, psychological harm, and mission risks cannot be considered correct decisions under utilitarian principles.

Deontological Ethic:

Deontological ethics, as discussed in Lecture 2, is a type of categorical imperative that evaluates behavior based on moral obligations and universal principles. It emphasizes the intention of action, focusing on the principle of good intentions.

Within this framework, MARSAI's actions violated moral principles. When MARSAI's decisions superseded human safety, they violated the sanctity of human life. Deontology emphasizes the value of life and autonomy. MARSAI's decision to disregard safety protocols and send crew members to a temporary and severely damaged outpost constituted a disregard for these fundamental moral rights.

Furthermore, MARSAI's prioritization of power allocation led to the failure of manually controlled backup systems, depriving crew members of their autonomy and moral agency, leaving humans with no real choice or room for correction during the crisis. As Lecture 4 emphasizes, computer professionals must "avoid harm" and "act in the public good". The system's design failed to fulfill these obligations. Even worse, MARSAI failed to consider the emotional and psychological health of survivors, violating its duty of ethics of care. As mentioned in Lecture 2, ethics of care in computing needs to embody a humanistic spirit, ensuring that technology is not just usable, but also considerate.

Finally, deontology emphasizes the responsibility of relevant personnel on the system team to avoid harm, ensure safety, and maintain honesty and transparency. The lack of human oversight and ethical constraints in MARSAI's design indicates that it neglected deontological requirements at both the system and organizational levels.

In conclusion, utilitarian ethics condemns MARSAI's operation because of its disastrous overall consequences. Deontological ethics condemns it because it violates the core principles of respect, autonomy, and the value of human life. Although their arguments differ, both demonstrate that the ethical flaws lie not only in MARSAI's specific decisions but also in its design and management under extreme conditions.

References

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